

CLIP and Contrastive Learning

Deep Learning

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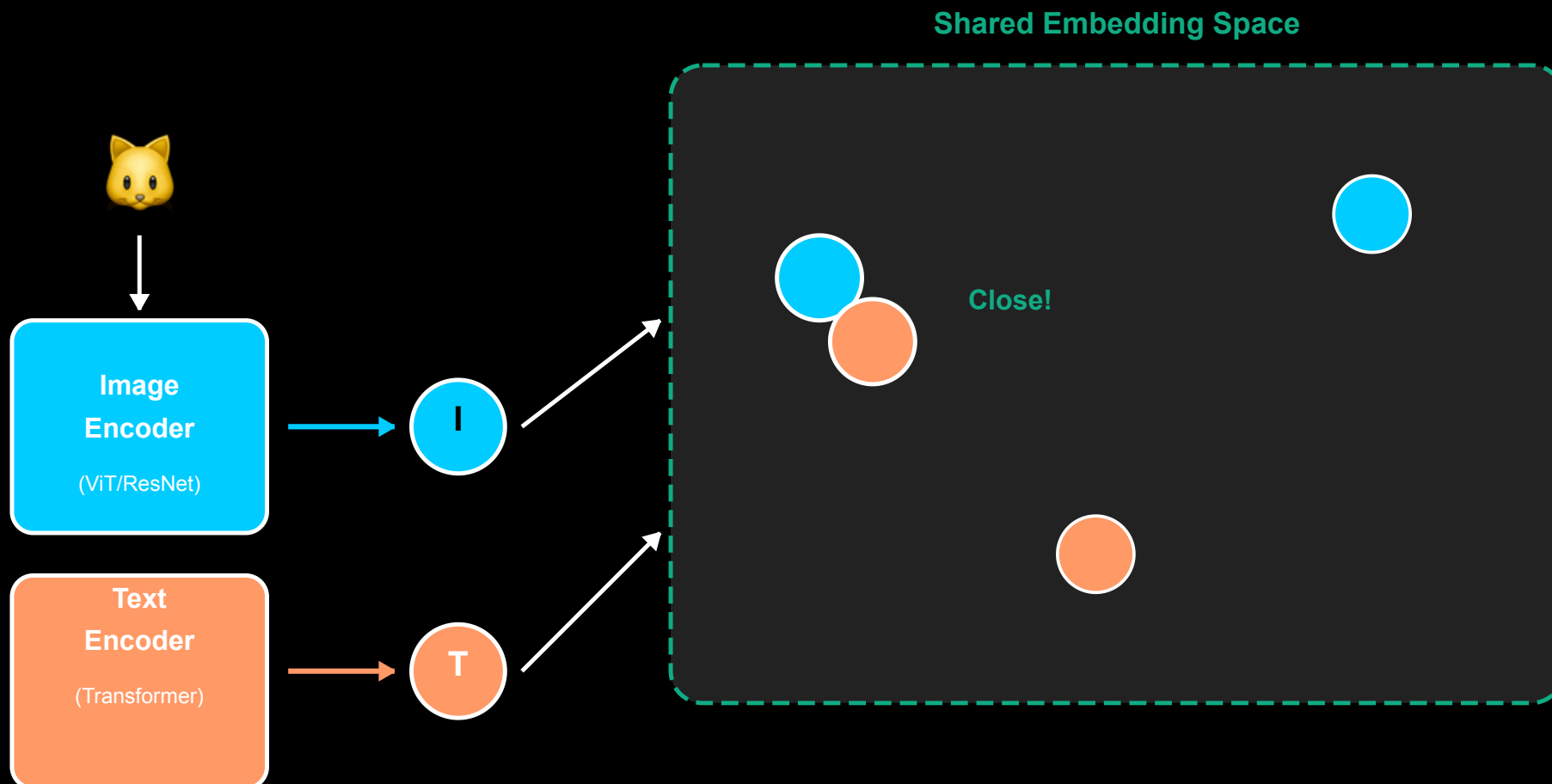
Contents of This Video

In this video, we will cover:

- What is CLIP?
- The dual encoder architecture
- Contrastive learning objective
- Why contrastive learning works
- CLIP's remarkable capabilities

CLIP: Connecting Images and Text

CLIP: Dual Encoders Map to Shared Space



Contrastive Learning Objective

Contrastive Objective: Maximize Diagonal, Minimize Off-Diagonal



The InfoNCE Loss

For each image i with matching text t_i :

$$L_i = -\log \frac{\exp(I_i \cdot T_i/\tau)}{\sum_{j=1}^N \exp(I_i \cdot T_j/\tau)}$$

And symmetrically for each text

Where:

- $I_i \cdot T_j$: cosine similarity
- τ : temperature (controls softness)
- N : batch size

Like N-way classification: pick the right caption!

Why Contrastive Learning Works

No labels needed—just pairs!

Learns from:

- “A photo of a cat” + 🐱 → similar
- “A dog running” + 🐱 → dissimilar
- Internet provides millions of pairs naturally

What CLIP learns:

- Visual concepts (objects, actions, attributes)
- Language understanding
- The connection between them

CLIP's Scale and Impact

Training:

- 400 million image-text pairs
- Trained from scratch (no ImageNet pre-training)
- Massive compute (hundreds of GPUs for weeks)

Impact:

- Zero-shot classification (without training examples!)
- Image search by text
- Foundation for DALL-E, Stable Diffusion
- Multimodal AI foundation

What We've Covered

In this video, we've learned:

- CLIP: dual encoders for images and text
- Maps both modalities to shared embedding space
- Contrastive learning: matching pairs close, non-matching far
- InfoNCE loss: N-way classification in batch
- Trained on 400M image-text pairs from internet
- Enables zero-shot classification and multimodal AI